Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program
(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**AI Robot Control Generation and Simulation System**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification 1)	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Advanced Robotics & Manufacturing		Artificial Intelligence	
	Medium Category	Small Category	Medium Category	Small Category
	Industrial Robot Control Systems	AI-Based Robot Programming	Artificial Intelligence Applications	Behavior Imitation A

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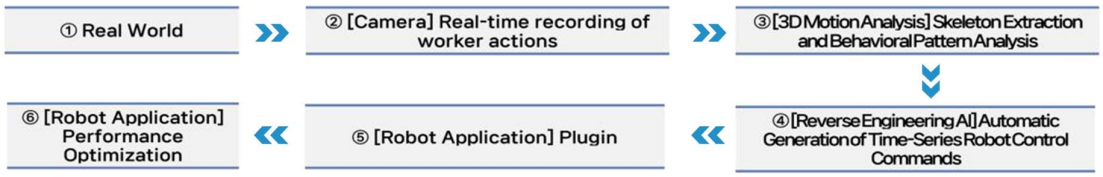
Concept of Proposed Technology	This project presents an AI-driven solution that upgrades existing industrial robots into autonomous systems through behavior imitation. Using a single camera, the system captures worker actions in real time, generates 3D motion avatars, performs AI-based reverse engineering, and automatically produces control code through a streamlined six-step process. This replaces the traditional 12-step programming workflow, reducing development time from six months to two weeks and lowering costs by 50–70%. The solution supports over 100 robot models, operates fully offline using edge AI, and incorporates digital twin simulation for validation and over-the-air (OTA) deployment. 
Rationale for Joint R&D Project	Traditional robot programming is complex, time-consuming, and costly, which limits the adoption of smart factory technologies among SMEs. A joint R&D approach is required in collaboration with Korean SMEs is essential to validate the proposed technology in real manufacturing environments, tailor it to specific industries (e.g., semiconductors and automotive), conduct joint proof-of-concept studies, and accelerate commercialization through shared expertise and market access. On the academic side, the University of Toronto team brings complementary expertise spanning core AI methods and embodied robotic systems. The effort is jointly led by Prof. Peyman Mohajerin Esfahani, who contributes in robust AI, behavior imitation learning, and data-efficient model design for real-world deployment, and Prof. Pakpong Chirarattananon, who leads research in robotics, manipulation, autonomous robot control, and rigorous experimental evaluation in physical environments. Together, the team integrates advances in imitation learning, 3D motion recognition, and edge AI–based simulation with hands-on validation on robotic platforms, enabling the development of reliable, adaptive control frameworks suitable for smart factory upgrades. Their combined strengths ensure both methodological innovation and system-level validation, forming a strong academic foundation for translation to SME manufacturing settings such as semiconductor and automotive production.
Key Words	1. Behavior Imitation AI

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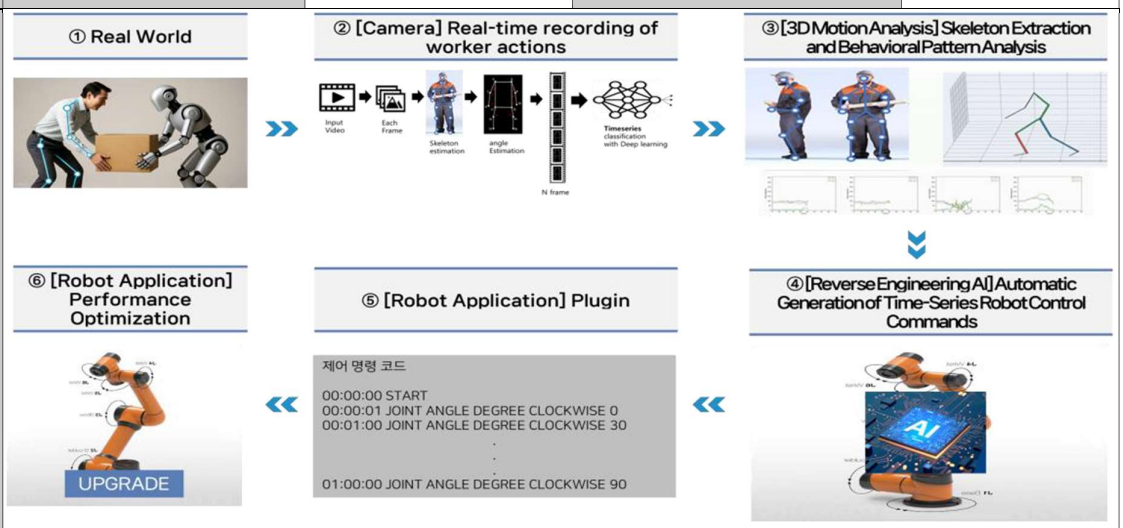
	2. Autonomous Robot Control 3. 3D Motion Recognition 4. Smart Factory Upgrade 5. Edge AI Simulation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives



The project will develop an integrated AI system for real-time, single-camera 3D beh avior recognition, AI-based reverse engineering for automatic robot control code gene ration, and digital twin–based simulation for validation. It will further expand compatibi lity to over 100 robot models and optimize offline edge AI operation for deployment i n indus trial environments.

The following are the main objective of these projects:

1. Develop an AI-based robot control simulator that learns diverse task behaviors through behavior imitation learning.
2. Deploy and validate simulated control solutions on physical robots and industri al controllers in real production settings.
3. Design and integrate camera and LiDAR modules to enable advanced 3D mo tion and behavior recognition.
4. Evaluate system performance and transferability using real-world 3D robot mo dels and manufacturing scenarios.

R&D Target Specifications

Category	Metric / Performance Indicator	Specification	Improvement / Advantage
Behavior Imitation	Real-time 3D action recognition accuracy	95%+	High-precision imitation without wearables
Robot Trajectory	Trajectory reproduction error (end-effector position)	<5mm	Sub-millimeter precision
	Collision avoidance success rate	95%+ (real-time Jacobian-based avoidance)	Safe multi-robot / human collaboration
	Speed & acceleration profile optimization	Optimized dynamic profile	Smoother, energy-efficient motion
Robot Control	Control code generation time	Minutes (automatic generation)	99% time reduction
	Robot model compatibility	100+ models	Universal plug-and-play
	OTA deployment & update success rate	100% success via OTA	Zero downtime for updates

R&D Plan

Phase 1 (Months 1–6): Improve behavior recognition using Lifting Network and LSTM; test in the lab.

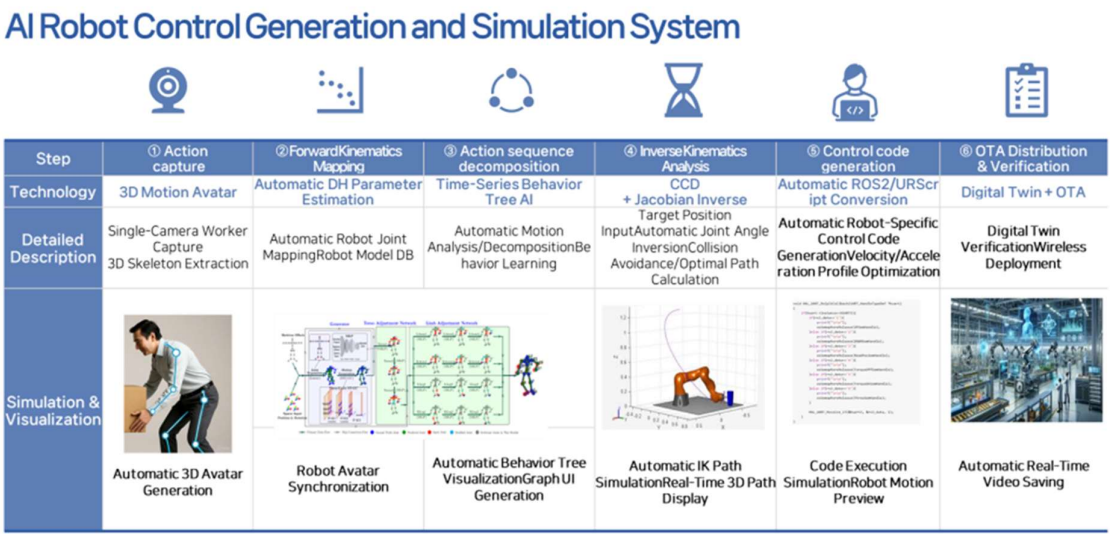
Phase 2 (Months 7–12): Add inverse kinematics and control code generation; build digital twin simulation.

Phase 3 (Months 13–18): Run proof-of-concept with SME partners in real factories; optimize edge AI for offline use.

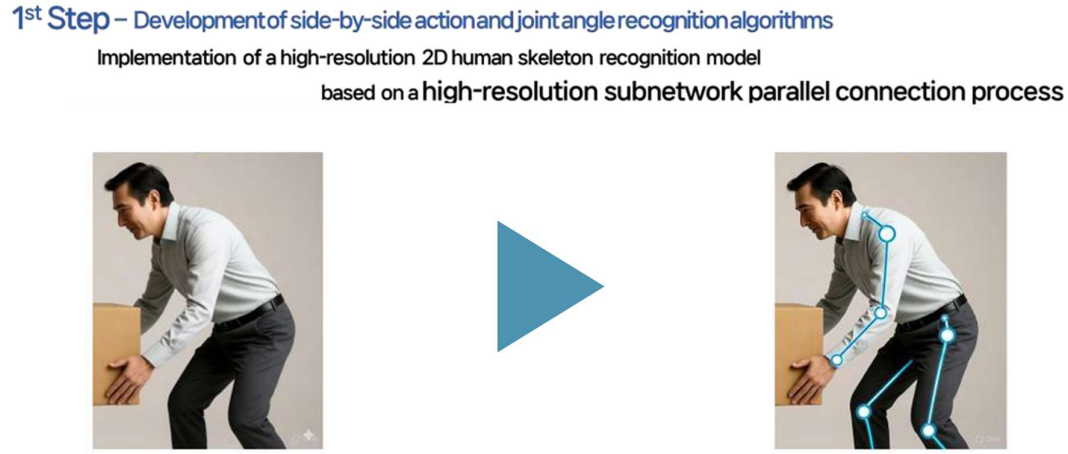
Phase 4 (Months 19–24): Finalize the system, complete certifications, and prepare for commercialization.

Milestones: Prototype v2.0 (Month 12), PoC success (Month 18), Market-ready system (Month 24).

The following figures capture the essence of the R&D plan, illustrating each phase, key activities, and expected milestones. They provide a clear roadmap from lab-scale validation to market-ready deployment.



For a more detailed and transparent explanation, we present the contents of each step.



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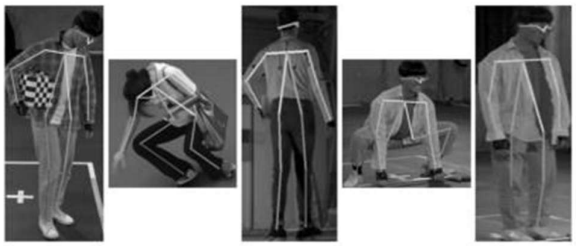
1st Step – Development of side-by-side action and joint angle recognition algorithms

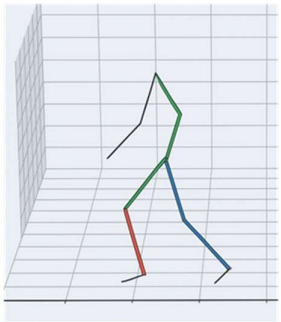
Implementation of an N-stage action recognition deep learning model



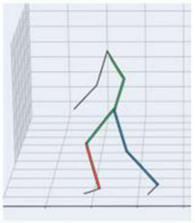
Moving Falling Walking Lifting Standing

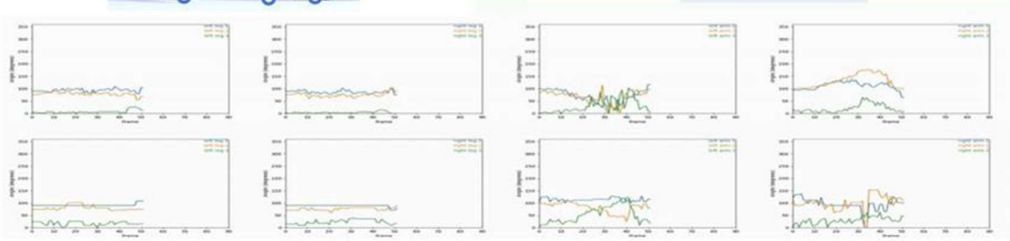
1st Step – Development of side-by-side action and joint angle recognition algorithms

Implementation of a 3D Human Skeleton Recognition Model



1st Step – Development of a side-by-side action and joint angle recognition algorithm





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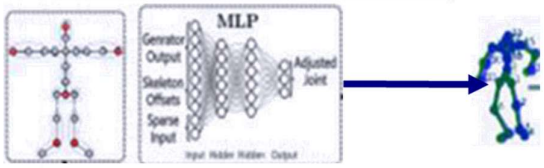
2nd Step – Forward Kinematics Mapping

Automatic DH parameter estimation, automatic robot joint mapping and robot model DB

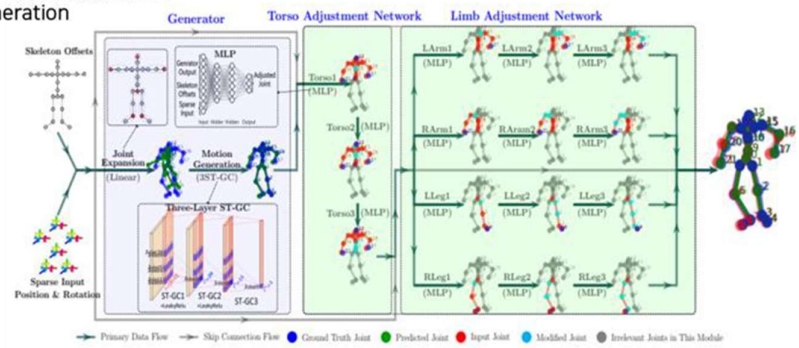
Robot joint synchronization



3rd Step - Action sequence decomposition

Time-series behavior tree AI
Automatic behavior analysis/decomposition, and behavior learning

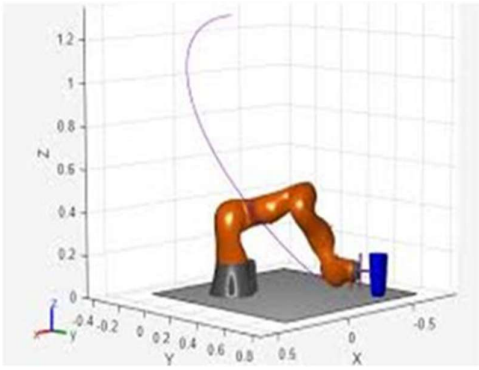
Automatic visualization of behavior trees
Graph UI generation



4th Step – Inverse Kinematics Calculation

CCD + Jacobian Inverse, Target position input joint angle automatic inversion collision avoidance + optimal path calculation

Automatic IK path simulation
Real-time 3D path display



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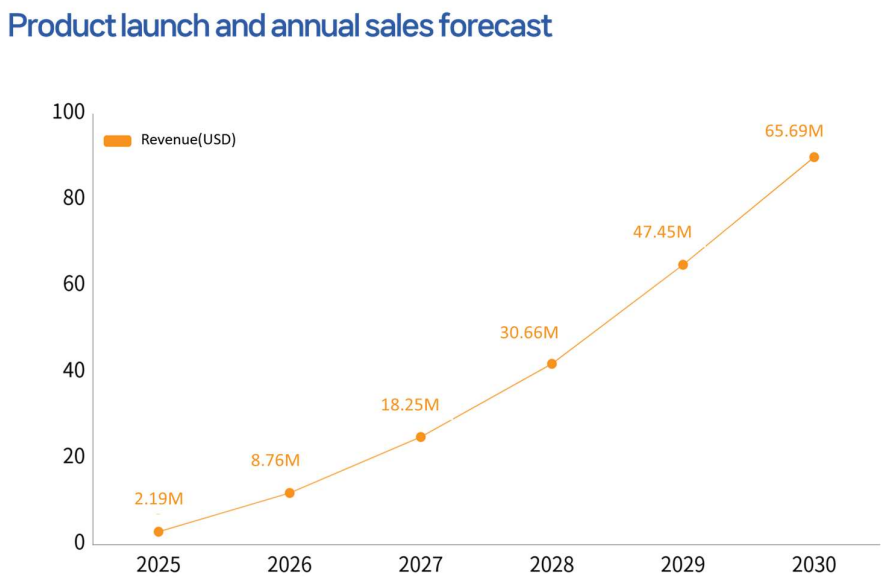
4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Provide advanced AI algorithms for behavior imitation and 3D motion analysis, conduct joint laboratory testing, offer technical expertise in robotics simulation, contribute to intellectual property development, and support customization for specific SME applications.
Expected R&R of Korean SME Partner	The Korean SME Partner will lead system integration, prototype development, and field testing at manufacturing sites, conduct market analysis to support commercialization, conduct proof-of-concept activities with end users, and be responsible for overall project management and financing coordination.

5. Expected Outcomes/Impact

Expected Outcomes and Impact



The project will deliver a commercial-ready AI robot control system, substantially reducing deployment time and costs for SMEs while enhancing Korea's smart manufacturing competitiveness. Projected revenue is $2.19M in 2025, $8.76M in 2026, $18.25M in 2027, $30.66M in 2028, $47.45M in 2029, and $65.69M in 2030. Economic impact includes cumulative sales of approximately $172.8M by 2030, the creation of 30 jobs, and ESG contributions through robot reuse, improving energy efficiency by 30%. Broader industrial effects include an improved global market share in AI robotics (15% by 2030) and technology transfer to SMEs.

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6. References

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(2) Xin Yu Kong et al., "Hierarchical Model-Based Imitation Learning for Planning in Autonomous Driving", arXiv preprint arXiv:2210.09539, 2022, https://arxiv.org/pdf/2210.09539

(3) World Economic Forum, "Physical AI: Powering the New Age of Industrial Operations", 2025, https://reports.weforum.org/docs/WEF_Physical_AI_Powering_the_New_Age_of_Industrial_Operations_2025.pdf

(4) Stanford HAI, "Artificial Intelligence Index Report 2025", 2025, https://hai-production.s3.amazonaws.com/files/hai_ai_index_report_2025.pdf

(5) International Federation of Robotics, "World Robotics 2025", 2025, https://ifr.org/img/worldrobotics/Online_Flyer_World_Robotics_2025_small.pdf

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization